

Autonomous Vehicle Image Classification with Pixel Perturbation Attack and Defense

This project demonstrates how Machine Learning can be used in autonomous vehicle vision systems to recognize objects and analyze the impact of adversarial attacks on image classification.

The CIFAR-10 dataset is used as a simplified representation of real-world objects that autonomous vehicles may encounter. Instead of using deep CNN architectures, the project focuses on traditional Machine Learning techniques combined with feature engineering.

Technologies and Techniques Used

Feature Extraction

- HOG (Histogram of Oriented Gradients) is used to extract important shape and edge information from images.
- HOG helps identify visual patterns such as object boundaries, similar to how autonomous vehicles detect lanes, vehicles, pedestrians, and traffic signs.

Machine Learning Models

- Logistic Regression
- Support Vector Machine (SVM)
- Random Forest

These models learn from HOG features and classify images into different categories.

Pixel Perturbation Attacks

A major objective of the project is to evaluate how small image modifications can affect AI decision-making. These modifications are called pixel perturbations.

1. Gaussian Noise

- Random noise is added to image pixels.
- Simulates rain, fog, low-light conditions, and camera sensor noise.

2. Salt-and-Pepper Noise

- Random black and white pixels are inserted into images.
- Simulates transmission errors and camera malfunctions.

3. Pixel Attack

- Random pixels in the image are modified.
- Demonstrates how small pixel changes can confuse an AI model and lead to incorrect classifications.

Attack Analysis

For each attack, the project measures: • Prediction before attack • Prediction after attack • Confidence score changes • L2 distance (amount of image modification)

Visual comparisons are generated to show: • Original image • Attacked image • Perturbation map highlighting pixel changes

Defense Mechanisms

- Gaussian Blur – Smooths image noise before classification.
- Median Filtering – Removes isolated noisy pixels while preserving important image details.
- Anomaly Detection – Detects potential attacks by monitoring sudden drops in model confidence.

Performance Evaluation

The project evaluates model performance under different noise levels and compares: • Accuracy on clean images • Accuracy on attacked images

This helps measure how resilient each model is to image corruption and adversarial perturbations.

Neural Network Extension

An additional Multi-Layer Perceptron (MLP) neural network is implemented using PyTorch to explore deep learning-based classification and prepare the framework for future adversarial attack techniques such as FGSM (Fast Gradient Sign Method).

Relevance to Autonomous Vehicles

Autonomous vehicles rely heavily on camera-based perception systems to identify road objects and make driving decisions. Environmental conditions, sensor noise, or malicious adversarial attacks can alter image pixels and potentially cause misclassification. This project demonstrates how such pixel perturbations affect AI models and evaluates defense mechanisms that improve system reliability, safety, and robustness in autonomous driving environments.